



同济大学
TONGJI UNIVERSITY

2025 Conference on Empirical Methods in
Natural Language Processing
(EMNLP2025)

DatawiseAgent

面向端到端数据科学自动化的
Notebook-Centric LLM Agent

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背景与模型介绍

DatawiseAgent 论文背景、任务介绍

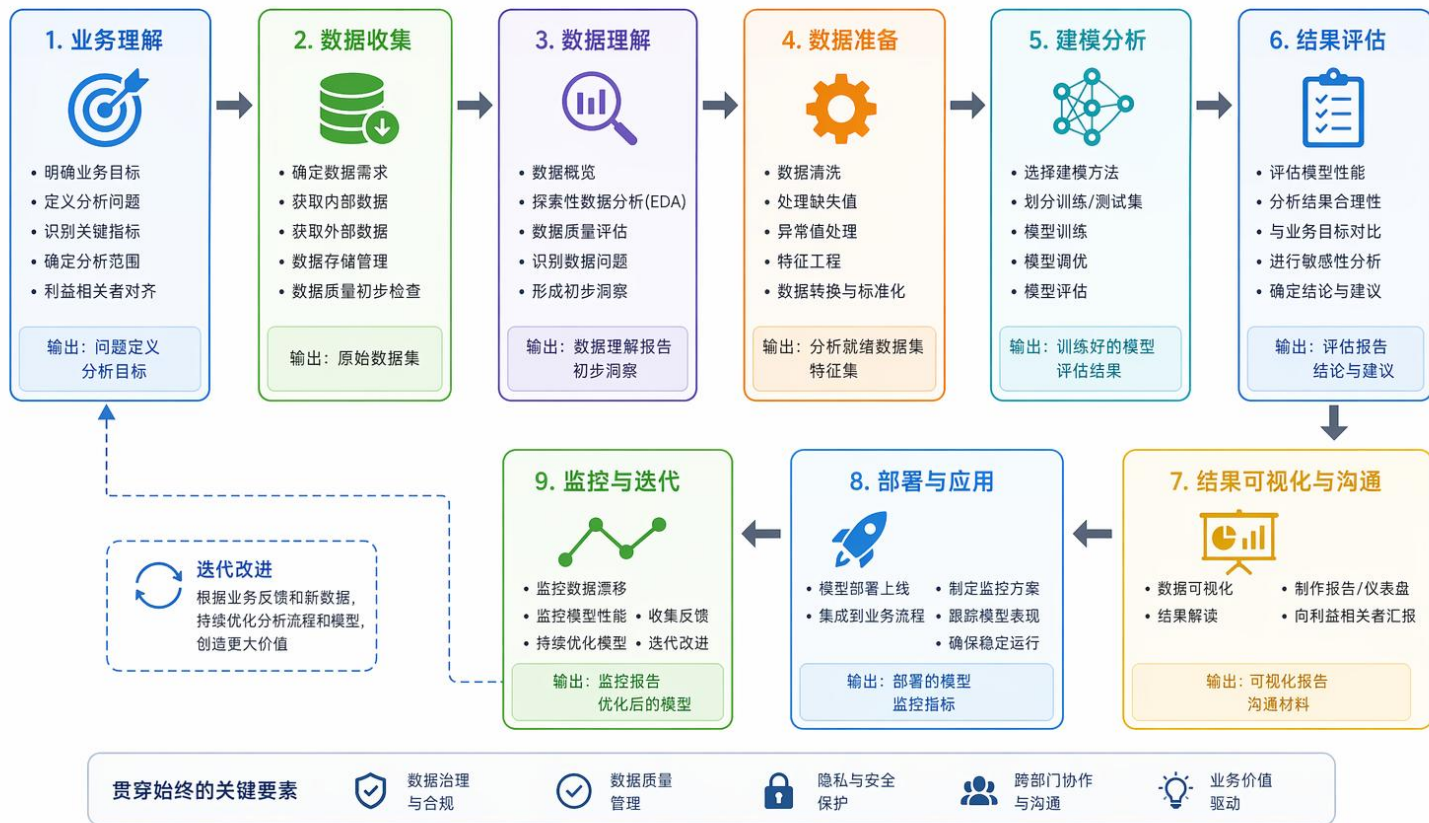


研究内容

- **目标：**让大语言模型完成完整的数据科学流程
- 不只回答问题，而是能够分析、编码、调试、建模、生成结果
- 更接近人类在 *Notebook* 中完成任务的真实工作方式

数据科学分析流程图

从业务理解到价值输出的端到端数据科学分析流程



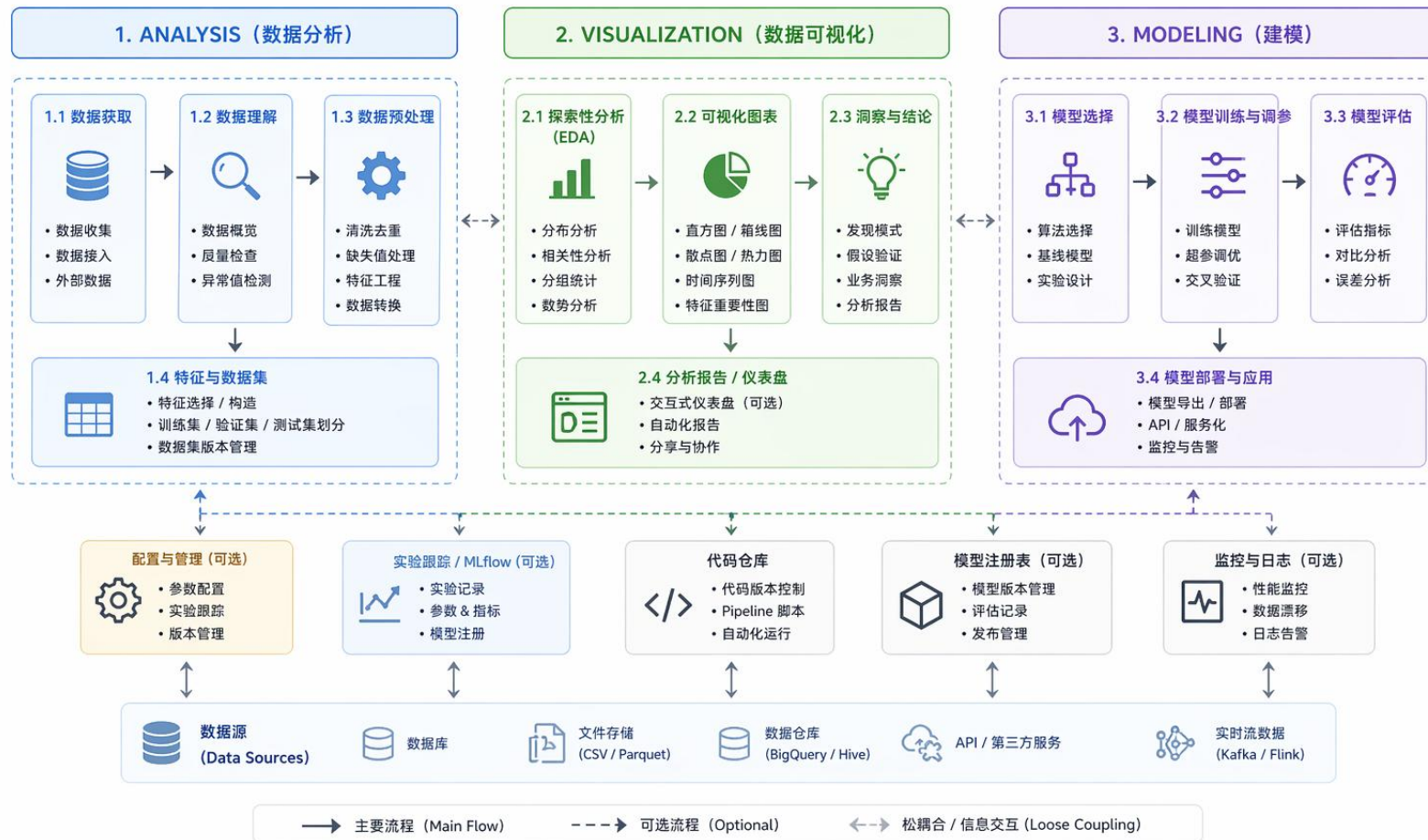
从“单点能力”转向“端到端流程能力”。



传统方法的局限

传统 ML 工作流程 Pipeline (Analysis / Visualization / Modeling 分离)

各模块职责清晰，松耦合、可独立迭代



现有方法缺陷:

- 任务割裂
- 泛化性差
- 对强模型依赖严重

传统 *AutoML* 只覆盖单一阶段, 例如模型选择或调参。现有 *LLM agent* 虽更灵活, 但在长链条任务中容易失控, 在完整数据科学流程上仍存在明显短板。一旦模型能力下降, 整体效果衰减明显



实验覆盖的三类典型任务

- 数据分析

Benchmark :
InfiAgentBench

内容：表格理解、统计分析、
特征解释

- 科学可视化

Benchmark : MatplotBench

内容：根据文本要求生成语义正确的
图表

- 预测建模

Benchmark : DataModeling /
DSBench

内容：读取训练集与测试集，完成
端到端建模并生成提交文件

sales_data_sample.csv

	A	B	C	D	E	F	G	H	I
1	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDERDATE	STATUS	CUSTOMERNAME	COUNTRY
2	10100	30	95.7	2	2871	2003-01-06	Shipped	Land of Toys Inc.	USA
3	10100	50	34.8	5	1740	2003-01-06	Shipped	Land of Toys Inc.	USA
4	10101	22	21.78	2	479.16	2003-01-09	Shipped	Reims Collectables	France
5	10101	49	48.86	6	2394.14	2003-01-09	Shipped	Reims Collectables	France
6	10102	25	99.23	2	2480.75	2003-01-10	Shipped	Toys4GrownUps.com	USA
7	10102	26	35.29	4	917.54	2003-01-10	Shipped	Toys4GrownUps.com	USA
8	10103	46	18.69	1	859.74	2003-01-29	Shipped	Corporate Gift Ideas Co.	USA
9	10103	45	45	2	2025.00	2003-01-29	Shipped	Corporate Gift Ideas Co.	USA
10	10104	39	18.69	1	728.91	2003-01-31	Shipped	Online Diecast Creations Co.	USA
11	10104	41	45	2	1845.00	2003-01-31	Shipped	Online Diecast Creations Co.	USA
12	10105	30	18	2	540.00	2003-02-11	Shipped	Tokyo Collectables, Ltd	Japan
13	10105	40	18	3	720.00	2003-02-11	Shipped	Tokyo Collectables, Ltd	Japan
14	10106	25	25.74	1	643.50	2003-02-17	Shipped	Auto-Associates & C.	USA
15	10106	26	49.64	2	1290.64	2003-02-17	Shipped	Auto-Associates & C.	USA
16	10107	38	100	1	3800.00	2003-02-24	Cancelled	Microscale Inc.	USA
17	10107	41	60	2	2460.00	2003-02-24	Cancelled	Microscale Inc.	USA
18	10108	30	76.69	1	2300.70	2003-02-26	Shipped	Signal Collectibles Ltd.	USA
19	10108	31	35	2	1085.00	2003-02-26	Shipped	Signal Collectibles Ltd.	USA
20	10109	49	55.09	1	2699.41	2003-03-03	Shipped	Classic Legends Inc.	USA
21	10109	24	56.55	2	1357.20	2003-03-03	Shipped	Classic Legends Inc.	USA
22	10110	25	24.68	1	617.00	2003-03-10	Shipped	Handji Gifts & Co	USA
23	10110	20	24.68	2	493.60	2003-03-10	Shipped	Handji Gifts & Co	USA
24	10111	42	46	1	1932.00	2003-03-18	Shipped	Frau da Collezione	Italy
25	10111	30	46	2	1380.00	2003-03-18	Shipped	Frau da Collezione	Italy

可视化任务 (示例)

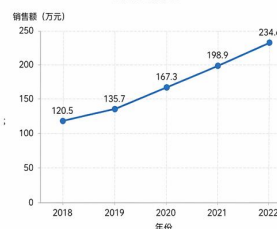
任务描述：
根据提供的电商订单数据（包含订单日期、
产品类别、销售额、利润、地区、子类别
等字段），完成以下可视化：

- 绘制 2018-2022 年各年销售额的折线图；
- 绘制各产品类别的销售额（柱状图），按销售额降序排列；
- 绘制各地区销售额的地图（中国地图，按销售额着色）；
- 绘制销售额与利润的散点图（点大小表示利润，颜色表示产品类别）。

请生成一个包含以上 4 个子图的图表。

Ground Truth Plot (参考答案)

1. 2018-2022 年各年销售额



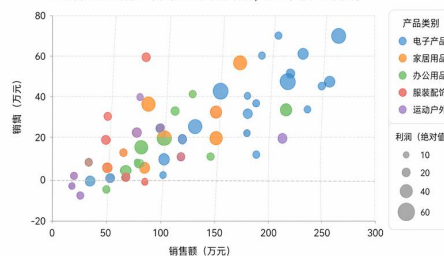
2. 各产品类别的销售额 (降序)



3. 各地区销售额地图 (中国地图)



4. 销售额与利润散点图 (点大小表示利润, 颜色表示产品类别)



注：以上图表为参考答案 (Ground Truth)，实际任务中数据和数值可能不同，但图表类型、结构和表达的同质保持一致。

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Overview Data Code Models Discussion Leaderboard Rules Submit Predictions

Submit Predictions

You can submit your predictions for this competition through the form below. Make sure to follow the type and format described in the Evaluation tab.

Submission Deadline
2 months to go
Fri, 1 Nov 2024 11:59:00 UTC

Submit File

Upload a CSV file with your predictions.

Download Sample Submission File

Drag and drop file here
Limit: 20MB per file • CSV

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Sample Submission

This is a sample submission file in the correct format.

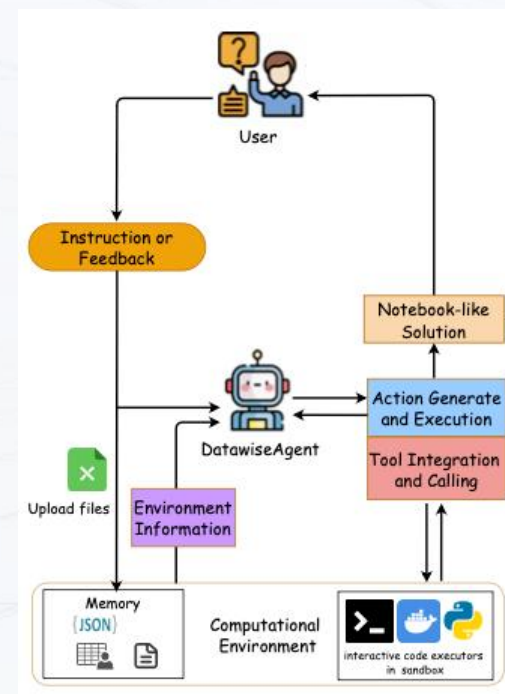
Id	SalePrice
1	208500.0
2	181500.0
3	223500.0
4	140000.0
5	250000.0
...	...
1461	175000.0

Rows: 1461 Columns: 2

这三类任务基本覆盖了数据科学从分析、可视化到建模的核心流程。

方法与分析

模型核心设计与分析

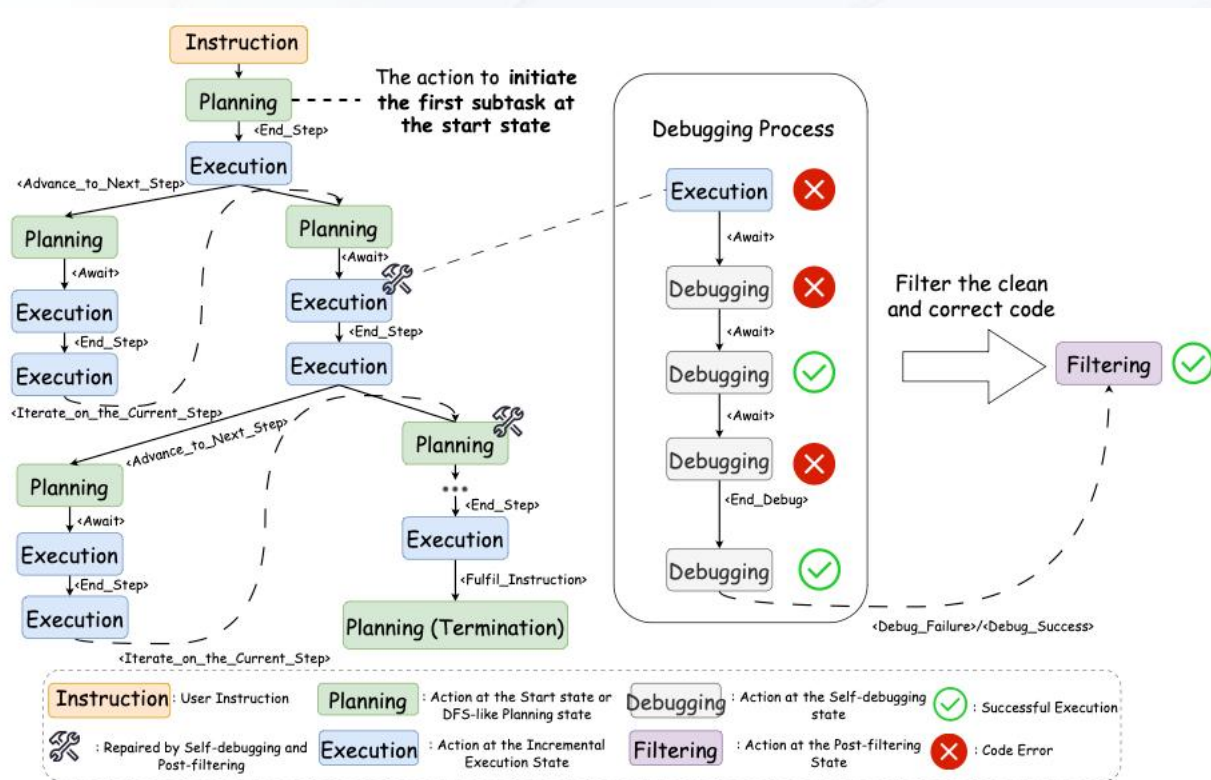


能写代码还不够，还需：

- 统一的 *Notebook* 表示
- 显式的多阶段状态控制

目标是让 *agent* 的行为更像人类在 *Jupyter* 中工作的过程

“规划—执行—报错—修复—筛选”循环，而不要一次性生成答案





核心设计一：Notebook-centric 统一表示

- 将用户指令、代码、执行结果、工具调用统一组织为 *notebook cell* 序列
- 用 “*Markdown + Code Cell*” 表达 *agent* 的完整推理过程
- 让推理过程具备可执行、可调试、可追踪的特点

Agent Trajectory

Development Environment Overview Markdown

The development environment is managed using Conda with Python 3.10 as the interpreter with some basic libraries for data analysis and machine learning installed ... If additional packages are required, install them directly using `!pip install` as needed.

`# show the packages installed` Executable Code

```
! pip show numpy pandas matplotlib
! pip show scikit-learn | grep Version
# show the accessible file system
! tree -F ./
```

...

Code Output

`# You can use the function 'read_image()' to access the descriptive and analytical information about any image files.` Executable Code

```
from system.vision_tool import read_image
import inspect
signature = inspect.signature(read_image)
print(signature)
print(read_image.__doc__)
```

...

Code Output

 **baro_2015.csv** Markdown

[USER INSTRUCTION]: Question: 1. What is the mean wind speed in the dataset? Calculate the arithmetic mean of WINDSPEED, excluding any null values. The mean must be calculated to three decimal places. baro_2015.csv has been uploaded. 2. ...

[STEP GOAL]: Load the dataset from the baro_2015.csv file and calculate the arithmetic mean of the WINDSPEED column, excluding any null values. The mean should be calculated to three decimal places ... Markdown

...

`# Load the dataset from the input directory` Executable Code

```
file_path = './input/baro_2015.csv'
data = pd.read_csv(file_path)
```

`# Display the first few rows to understand its structure` Code Output

```
data.head()
```

	DATE TIME	WINDSPEED	DIR	GUSTS	AT	BARO
0	01/01/2015 00:00	2.72	208	5.25	25.7	1023.0
1	01/01/2015 01:00	3.89	273	7.00	26.6	1022.7
2	01/01/2015 02:00	4.86	268	6.41	27.0	1022.1
3	01/01/2015 03:00	4.47	294	7.10	26.6	1021.4
4	01/01/2015 04:00	4.98	293	7.19	25.9	1020.9

The debugging process identified that ... Markdown

which caused a `KeyError`. By ... , we were able to correctly access the `WINDSPEED` column and calculate the mean wind speed. Below is the complete and corrected code to load the dataset and compute the mean wind speed.

`# Load the dataset from the input directory` Executable Code

```
file_path = './input/baro_2015.csv'
data = pd.read_csv(file_path)
# Strip whitespace from column names
data.columns = data.columns.str.strip()
```

`# Calculate the mean of WINDSPEED, excluding null values` Code Output

```
mean_windspeed = data["WINDSPEED"].mean()
# Print the mean wind speed rounded to three decimal places
print(f"Mean Wind Speed: {mean_windspeed:.3f}")
```

Mean Wind Speed: 5.979

The task of calculating the mean wind speed from the dataset baro_2015.csv has been successfully completed. Markdown

The arithmetic mean of the `WINDSPEED` column, excluding any null values, is **5.979 m/s**, rounded to three decimal places.

If you have any further questions or need additional analysis, feel free to ask!

优点

- 上下文结构统一
- 更贴近人类 *Jupyter* 工作方式
- 执行反馈可直接进入下一轮推理



为何 Notebook 而非对话?

- 对话式 *agent* 更像“问答系统”
- Notebook 式 *agent* 更像“工作系统”
- 每一步都有显式中间状态：代码、输出、错误、修复痕迹
- 对长任务尤其重要，因为可以保留结构化执行历史

Chat 界面（对话式）

DataSci Assistant

请分析 train.csv 数据，目标是预测房价（SalePrice）。先帮我做数据概览，包括缺失值统计和数值特征的分布情况。

好的，我已经加载了 train.csv 数据，并进行了初步的数据概览分析。

✓ 数据概览

- 数据形状：1460 行 × 81 列
- 目标变量：SalePrice（数值型）
- 缺失值统计（前10列）：

列名	缺失数量	缺失比例
LotFrontage	259	17.74%
Alley	1369	93.77%
MasVnrType	8	0.55%
MasVnrArea	8	0.55%
BsmtQual	37	2.53%
...	...	...

• 数值特征分布（部分）

如需查看其他特征分布或相关性分析，请告诉我。

查看相关性热力图 按分组查看房价分布 继续特征工程建议

发送消息...

内容由 AI 生成，请仔细甄别

Notebook 界面（代码式）

house_price_ed.ipynb

File Edit View Insert Cell Kernel Widgets Help

Code Python 3 (ipykernel)

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

In [2]: df = pd.read_csv('train.csv')
df.shape

Out[2]: (1460, 81)

In [3]: # 缺失值统计（前10列）
df.isnull().sum().sort_values(ascending=False).head(10)

Out[3]: Alley          1369
aLotFrontage         259
MasVnrType             8
MasVnrArea             8
BsmtQual              37
BsmtCond              37
BsmtExposure          38
BsmtFinType2          38
Electrical              1
GarageType             1
dtype: int64

In [4]: # 数值特征分布（部分）
num_cols = ['SalePrice', 'GrLivArea', 'TotalBsmtSF', 'LotArea']
df[num_cols].hist(bins=30, figsize=(12, 3))
plt.tight_layout()

Out[4]:
```

```
In [ ]:
```



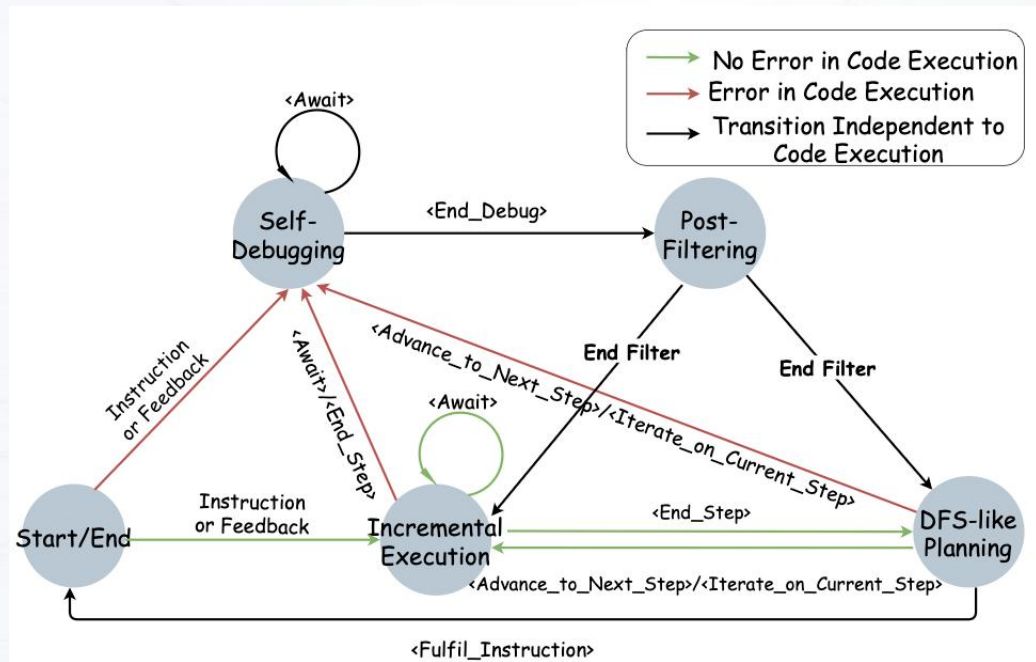
同一任务
不同交互方式

它不是在“聊天”，而是在“做事”。

核心设计二：基于 FST 的多阶段控制

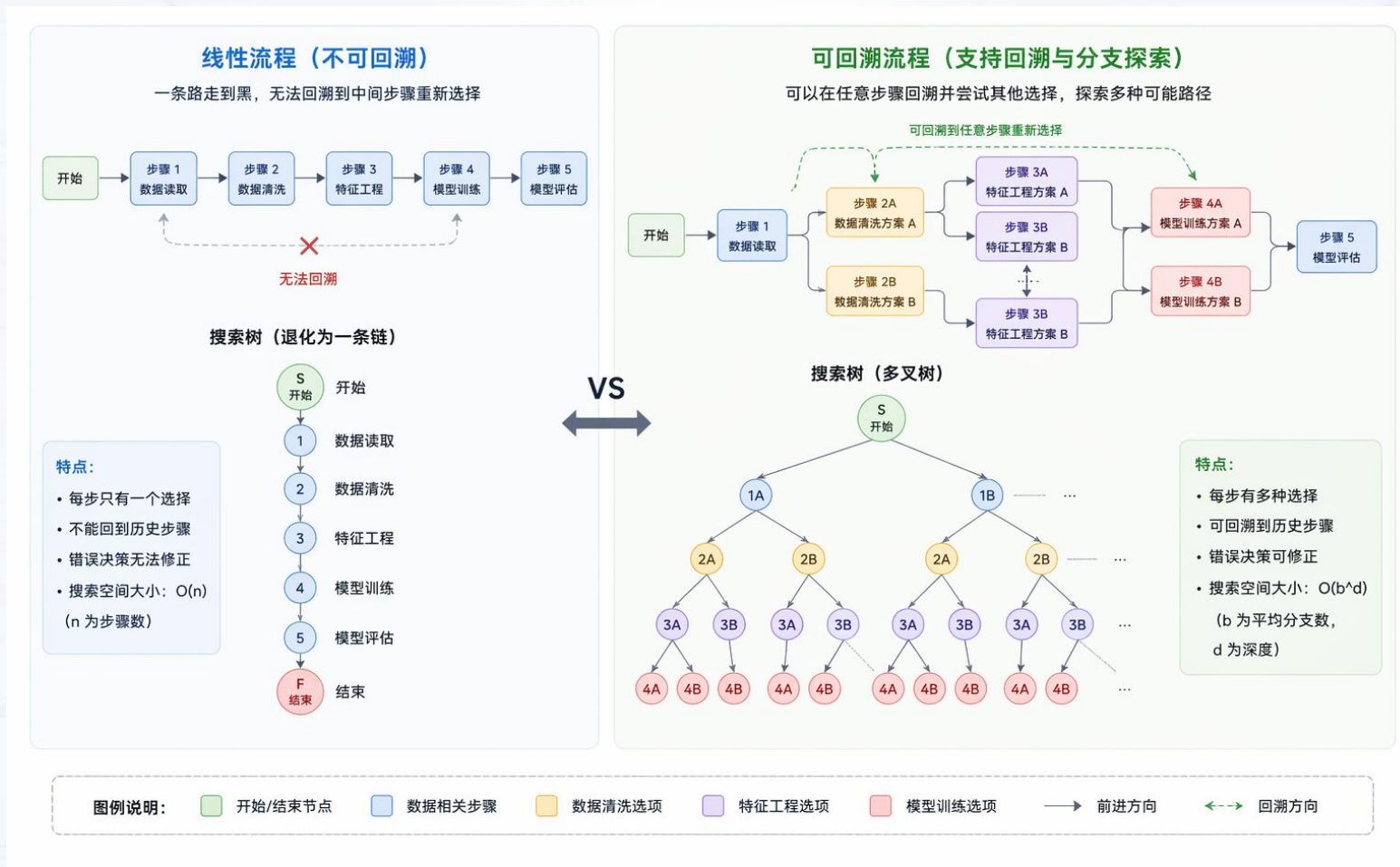
文本：四个关键阶段：

- *DFS-like Planning*
任务分解，
前进 / 回溯 / 重规划
- *Incremental Execution*
逐步执行，而不是一次性生成整段代码
- *Self-debugging*
根据报错自动修复
- *Post-filtering*
过滤无效或错误轨迹，
只保留有效执行结果
关键不是模块本身，而是模块之间可以来回跳转。



- 论文把 *agent* 的行为拆成多个阶段
- 更重要的是：阶段之间可以根据反馈动态跳转
- 因此它不是固定 *pipeline*，而是状态机驱动的执行过程

这种结构为什么有效?



- **长链条任务:** 不会因为一步失败而整段崩掉
- **非线性探索:** 允许类似搜索树的局部试错
- **对弱模型友好:** 可以靠多轮修正弥补单次生成能力不足

论文结果与观察

论文结果（示例）

Model	Method	ABQ/% ↑
GPT-4o mini	ReAct	80.08
	AutoGen	70.04
	Taskweaver	76.65
	Data Interpreter	67.70
	DatawiseAgent (Ours)	82.88
GPT-4o	ReAct	81.32
	AutoGen	73.54
	Taskweaver	85.99
	Data Interpreter*	94.93*
	Data Interpreter	75.78
Qwen2.5-72B-Instruct	ReAct	75.88
	AutoGen	70.04
	Taskweaver	74.71
	DatawiseAgent (Ours)	81.71

Table 1: Performance comparison on InfiAgent-DABench. Asterisk (*) result is from Hong et al. (2024) and could not be reproduced in our setting; it is

Framework	Model	Task Success/%	RPG
AutoGen	Llama3-8B	5.41	1.55
	Llama3-70B	16.22	7.79
	GPT-4	87.84	45.52
	GPT-4o	71.62	34.74
	GPT-4o mini	22.97	11.24
Code Interpreter	GPT-4	54.05	26.14
	GPT-4o	44.59	19.87
	GPT-4o mini	39.19	16.90
Human*	Human*	100.00*	65.02*
DatawiseAgent	GPT-4o	98.64	53.18
	GPT-4o mini	98.64	46.61
	Qwen2.5-72B	91.89	42.90

Table 3: Performance comparison on 74 Data Modeling tasks from DSbench. Bold and underline indicate

Model	Framework	Avg. Score ↑	Δ Score ↑
GPT-4o mini	Direct Decoding	38.09	-
	MatplotAgent	51.44	+13.35
	AutoGen	51.82	+13.73
	w/ visual tool	52.07	+13.98
	DatawiseAgent w/ visual tool	55.85	+17.76
GPT-4o	Direct Decoding	45.28	-
	MatplotAgent	57.86	+12.58
	AutoGen	60.42	+15.14
	w/ visual tool	63.60	+18.32
	DatawiseAgent w/ visual tool	61.22	+15.94
Qwen2.5-72B-Instruct	Direct Decoding	47.54	-
	AutoGen	40.80	-6.74
	w/ visual tool	53.72	+6.18
	DatawiseAgent	56.41	+8.87
	w/ visual tool	61.88	+14.34

Table 2: Performance comparison on MatplotBench. Δ Score denotes the score gain over Direct Decoding.

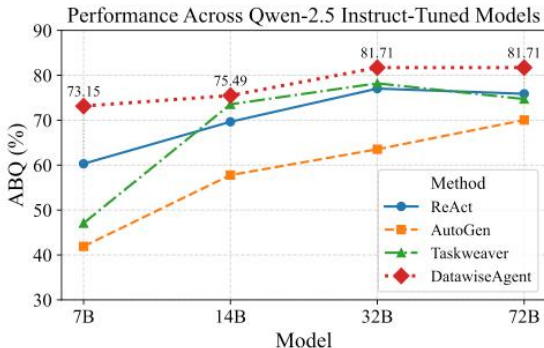


Figure 5: Performance across Qwen2.5 models on InfiAgent-DABench. DatawiseAgent demonstrates

分析

Benchmark	论文指标	备注
InfiAgentBench	ABQ	验证数据分析问答的正确率，多模型下表现稳定
MatplotBench	平均得分	验证科学可视化结果质量；利用外部反馈迭代修图
DataModeling	Task Completion / RPG	验证End2End建模任务完成能力；

- 论文在三类 benchmark 上都展示了较强表现
- 从结果趋势看，优势主要来自“结构设计”而不只是模型规模
- 尤其在需要连续执行、调试和修正的任务中，DatawiseAgent 更稳定
- 这也解释了它为什么对弱模型更友好

它的收益来自 长流程规划 + 调试恢复 + 迭代优化执行。

复现

baseline

MatplotBench + InfiAgentBench + DataModeling



复现工作划分

复现分成三个 *benchmark*:

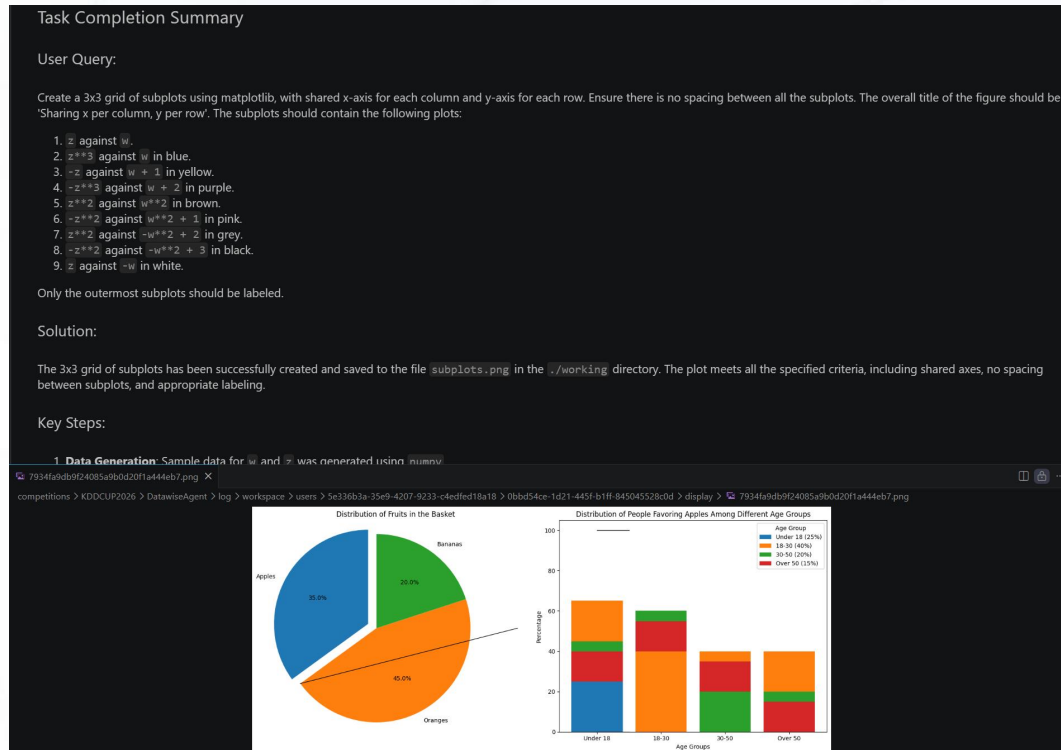
- *MatplotBench*
关注文本到图表的可视化生成能力
- *InfiAgentBench*
关注数据分析与表格推理能力
- *DataModeling*
关注类似 *Kaggle* 的端到端建模能力





MatplotBench

- *MatplotBench* 对应的是科学可视化任务，重点考察 *agent* 是否能把自然语言需求转成符合语义的图表。
- 这类任务不仅要求代码能运行，还要求图的布局、标注和视觉语义正确。
- 复现重点通常在于：图像生成是否符合要求，失败案例是否集中在标注错误、配色错误或布局偏差。
- 这一部分说明 *DatawiseAgent* 不仅会写代码，关键是能完成“可执行 + 可解释 + 可视化正确”的完整流程。



- *InfiAgentBench* 对应的是数据分析任务，主要测试 *agent* 对表格、统计和推理问题的处理能力。
- 相比可视化任务，这里更关注多步分析链条是否正确，例如数据读取、统计计算、条件筛选和结论表达。
- 这一部分能体现 *DatawiseAgent* 的强项：它不是只生成一段代码，而是能在执行、报错、修复之间反复迭代。
- 如果结果较好，可以强调它在长链条分析任务上的稳定性；如果有失败案例，可以指出复杂推理仍然依赖模型能力。

Question: Calculate the mean value of the "Close Price" column.
Use the built-in Python (numpy or pandas) to calculate the mean. Do not use any pre-built packages or libraries for mean calculation other than numpy or pandas. The calculation should be done on the whole "Close Price" column. Values in this column should not be rounded or changed in any way before the calculation.

Expected Format

@mean_close_price[mean_value], where "mean_value" is a float number rounded to two decimal places. This value should be between the highest and lowest "Close Price" given in the dataset.

[STEP GOAL]: Load the CSV file containing the "Close Price" column and calculate the mean value of that column using pandas.

First, we need to load the CSV file into a pandas DataFrame and then calculate the mean of the "Close Price" column.

```
import pandas as pd

# Load the dataset
df = pd.read_csv('./input/GOODRE3IND.csv')

# Calculate the mean of the "Close Price" column
mean_close_price = df["Close Price"].mean()

mean_close_price
```

Python

Time consumed: 0.06s
Code output:
Exit code: 0
Output:
570.6812753036437

The mean value of the "Close Price" column in the dataset is approximately 570.68.

[STEP GOAL]: Provide a detailed summary of the calculated mean value of the "Close Price" column.

The mean value of the "Close Price" column in the dataset `GOODRE3IND.csv` has been successfully calculated using pandas. The mean value is approximately **570.6812753036437**. This represents the average closing price across all entries in the dataset.

The mean value of the "Close Price" column in the dataset `GOODRE3IND.csv` has been successfully calculated. The average closing price across all entries in the dataset is approximately **570.6812753036437**. This value provides a central tendency measure for the close prices observed in the dataset.

The task to calculate the mean value of the "Close Price" column in the `GOODRE3IND.csv` dataset has been completed. The mean value was computed using pandas and determined to be approximately **570.6812753036437**. This value represents the average closing price across all entries in the dataset without rounding or altering the values in the "Close Price"



DataModeling

- 该任务要求 *agent* 完成数据读取、预处理、特征工程、训练脚本生成、预测和提交文件保存。
- 难点：运行链条长、环境依赖多、执行时间长、会话易中断。
- 一共24道 *DataModeling* 任务，完成率 $22/24 = 91.7\%$ ，汇总分数为 $performance = 0.293$ ，对 *benchmark* 数据完整性敏感。

```
!grep -E '(NAME|VERSION|ID|ID_LIKE)' /etc/os-release

# Get system information (CPU, GPU, Memory)
!echo "CPU Cores: $(lscpu | grep '^CPU(s):' | awk '{print $2}')"
!if command -v nvidia-smi >/dev/null 2>&1; then nvidia-smi --query-gpu=memory.total,memo
!echo "Memory Usage: $(free -h | grep Mem | awk '{print $2 " total, " $3 " used, " $4 "

import pandas as pd
# Disable HTML representation of dataframe
pd.options.display.notebook_repr_html = False
```

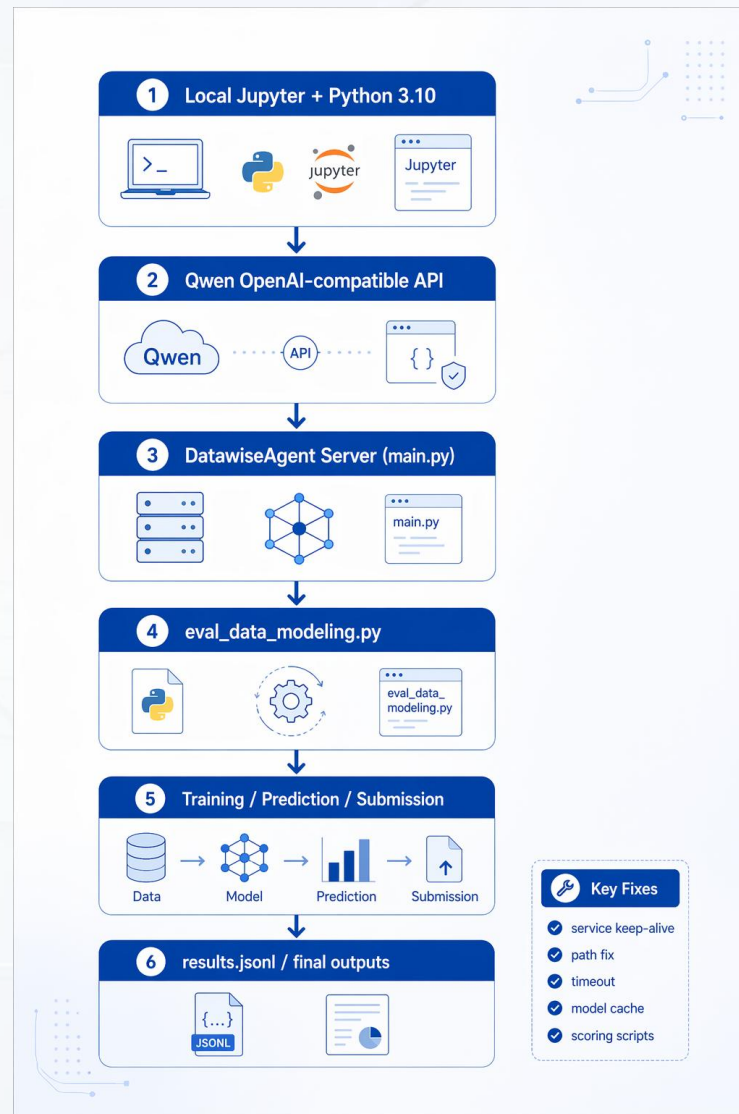
Problem	Fix
Sleep interruption	caffeinate keep-awake
Stale main.py port	kill stale process
6s timeout too short	extend to 20s
stale model cache / feature mismatch	clear stale model cache
score4each invalid	repair interpreter / script path

问题修复

Evaluable tasks:
14/24 → 22/24
Remaining 2 cases are due to missing real benchmark data (Git LFS pointer), not agent failure.

```
(.venv310) ivy@ivydeMacBook-Pro evaluation % ../venv310/bin/python DataModeling/scripts/show_res
74
Using model: qwen-dm-run7-autoretrain
Ground truth path: ./DataModeling/save_performance/GT
Output dir: ./experimental_results/Datamodeling-DSBench/qwen-dm-run7-autoretrain/performances/
Results path: ./experimental_results/Datamodeling-DSBench/qwen-dm-run7-autoretrain/results.jsonl
Task completion rate is 22/24=0.9166666666666666
The average time consuming is 1045.307187596957
The performance is 0.29298344591701325 with 24 out of 74 tasks scored.
```

结果示意



流程示意



改进方向思考

从三种**baseline**中的实际案例分析得到当前模型的缺陷，从而我们确立了改进方向

1 Matplot-bench

- 视觉反馈闭环要补本地 **fallback**
- 复杂 *Matplotlib* 场景需要专门的模板和单元测试
- *Notebook* 产物太啰嗦，导致日志膨胀
- 调试流程需要更强的错误归因

2 Data-modeling

- Git LFS 穿透导致的幻觉与死循环崩溃->引入“数据形态嗅探器”
- 高频的目标维度校验失败与数据穿越->利用代码生成图验证
- 算法搜索空间贫乏，重度依赖基线范式->“探索与利用”的 *Sub-agent* 框架
- 无视分布异常，鲁棒性缺失的粗暴缺失值填补->在主建模提示词中强制设立一个条件断言规则池

3 Infiagent-bench

- 长上下文中的强约束遗忘->将代码生成执行与最终结果按格式提取解耦
- 严重违背统计学原理的数据泄露
- 过度拟合成功执行，缺乏统计量置信度校验->防护性检查
- 静默崩溃导致上下文彻底丢失->引入重试降级与安全熔断机制

优化重点：控制 > 生成，策略 > 套路，积累 > 临场修补。

arXiv
> cs > arXiv:2503.07044

Computer Science > Computation and Language

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DatawiseAgent: A Notebook-Centric LLM Agent Framework for Adaptive and Robust Data Science Automation

Ziming You, Yumiao Zhang, Dexuan Xu, Yiwei Lou, Yandong Yan, Wei Wang, Huaming Zhang, Yu Huang

Existing large language model (LLM) agents for automating data science show promise, but they remain constrained by narrow task scopes, limited generalization across tasks and models, and over-reliance on state-of-the-art (SOTA) LLMs. We introduce DatawiseAgent, a notebook-centric LLM agent framework for adaptive and robust data science automation. Inspired by how human data scientists work in computational notebooks, DatawiseAgent introduces a unified interaction representation and a multi-stage architecture based on finite-state transducers (FSTs). This design enables flexible long-horizon planning, progressive solution development, and robust recovery from execution failures. Extensive experiments across diverse data science scenarios and models show that DatawiseAgent consistently achieves SOTA performance by surpassing strong baselines such as AutoGen and TaskWeaver, demonstrating superior effectiveness and adaptability. Further evaluations reveal graceful performance degradation under weaker or smaller models, underscoring the robustness and scalability.

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DatawiseAgent 的核心贡献是结合了：

- 统一 *Notebook* 表示 + 显式状态多阶段状态控制
- 相比传统方法，它更适合端到端、长链条、可执行、可调试、可反复修正的数据科学任务
- 我们的复现结果说明：框架主链路可以跑通，但 *DataModeling* 对数据完整性与执行环境稳定性要求更高

成员	负责内容
张恒	<i>MatplotBench</i> 复现与整理
於之翔	<i>InfiAgentBench</i> 复现与整理
冉子易	<i>DataModeling</i> 复现、调试与评分

谢谢！

张恒 於之翔 冉子易